

Crowd
state

Robot
observations

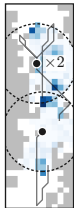
Reconstruction
methods

Output

Evaluation



x_t



y_t, Ω_t

 Senseiver-A	y_t
 Senseiver-G (ours)	$y_{t-15:t}$
 DINCAE*	$y_{t-1:t+1}$
 4DVarNet	200 s
 4DVarNet aug. var. (ours)*	200 s
 EnKF*	$y_{1:t}$

* also predicts its uncertainty $\hat{\sigma}_t$



$\hat{x}_t$



$\hat{\sigma}_t$

accuracy	RMSE
uncertainty	CRPS
	spread/RMSE
speed	ms per frame
on unobserved walkable cells	

ground truth (test days)